1. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

> *"Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation"*

- Highly relevant: This is the core of the solution process.
- Students must carry out the following logical steps:
  1. Recognize that "recovering the cost" means net gain $F \geq 0$;
  2. Substitute into the formula: $400{,}000y - 3{,}200{,}000 \geq 0$;
  3. Solve the inequality: $400{,}000y \geq 3{,}200{,}000$;
  4. Compute: $y \geq 3{,}200{,}000 \div 400{,}000 = 8$;
  5. Since the answer must be the smallest whole number of years, the result is 8 years (Option B).
- This constitutes a clear multi-step algebraic reasoning chain, fully aligned with Competency #12.

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2. Competency #3: Implement a given strategy to determine a mathematical solution

> *"Implement a given strategy to determine a mathematical solution"*

- Highly relevant: The problem provides a complete mathematical model (the formula), and students are expected to faithfully apply it.
- No modeling or method selection is required—students simply use the mayor's linear function as instructed.
- This is a textbook example of "implementing a given strategy," making Competency #3 central to the task.

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3. Competency #7: Manipulate numbers, algebraic expressions, equations, and other mathematical objects

> *"Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations..."*

- Strongly relevant: The solution involves key algebraic operations:
  - Performing division with large numbers: $3{,}200{,}000 \div 400{,}000$;
  - Working with a linear inequality (implied by "at least");
  - Simplifying the expression (e.g., canceling thousands: $3200 \div 400 = 8$).
- Although the arithmetic is straightforward, accurate manipulation of numbers and symbols is essential for a correct answer.

4. Competency #8: Articulate a solution, showing and/or summarising intermediate mathematical results

> *"Articulate a solution, showing and/or summarising and presenting intermediate mathematical results"*

- Moderately relevant: While the question is multiple-choice and does not require a written explanation, the internal reasoning must include intermediate steps (e.g., computing $y = 8$).
- In an open-response or instructional setting, students should be able to write:

> "Set $F = 0 : 400{,}000y = 3{,}200{,}000$, so $y = 8$."

- Thus, this competency is implicitly but necessarily involved in solving the problem.

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5. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process

> *"Use an understanding of the context to guide or expedite the mathematical solving process, e.g., working to a context-appropriate level of accuracy"*

- Moderately relevant: Contextual knowledge helps students:
  - Interpret "recover cost" as $F = 0$ (not $F > 0$);
  - Recognize that the number of years must be an integer, and they need the smallest integer satisfying the condition (so $y = 8$, not 7.999...);
  - Avoid unnecessary decimal precision or continuous-time modeling.
- This context-driven simplification makes the solution more efficient and realistic.

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6. Competency #1: Perform a simple calculation

> *"Perform a simple calculation"*

- Weakly but present: The final step requires only one division ($3.2\,\text{million} \div 0.4\,\text{million} = 8$).
- Though simple, this calculation is the concluding act of a multi-step process and represents basic execution skill.

1. 第12项：使用多步骤程序得出数学解、结论或一般化结果

> *"Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation"*

- 高度相关：这是解题的核心过程。
- 学生需执行以下逻辑步骤：
  1. 理解“收回成本”意味着 净收益 $F \geq 0$；
  2. 将条件代入公式： $400{,}000y - 3{,}200{,}000 \geq 0$；
  3. 解不等式： $400{,}000y \geq 3{,}200{,}000$；
  4. 计算： $y \geq 3{,}200{,}000 \div 400{,}000 = 8$；
  5. 因要求“最小整数年”，答案为 8 年（选项 B）。
- 这是一个清晰的多步代数推理链，完全契合第12项。

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2. 第3项：实施给定策略以确定数学解

> *"Implement a given strategy to determine a mathematical solution"*

- 高度相关：题目已提供完整数学模型（公式），学生只需忠实执行其应用。
- 无需自行建模或选择方法，而是直接使用市长给出的线性函数进行计算。
- 这正是“实施给定策略”的典型场景——将现实目标（盈亏平衡）转化为对公式的操作。

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3. 第7项：操作数字、代数表达式、方程等数学对象

> *"Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations..."*

- 强相关：解题涉及关键代数操作：
  - 处理大数（百万级）的除法： $3{,}200{,}000 \div 400{,}000$；
  - 操作线性不等式（隐含在“至少”中）；
  - 简化表达式（可约去单位“千”：3200 ÷ 400 = 8）。
- 虽然计算简单，但准确操作数值和符号是正确解答的基础。

4. 第8项：清晰表述解法，展示和/或总结中间数学结果

> *"Articulate a solution, showing and/or summarising and presenting intermediate mathematical results"*

- 中等相关：虽然题目为选择题，不要求书面过程，但内部推理必须包含中间步骤（如计算 $y = 8$ ）。
- 在教学或开放题情境下，学生应能写出：

> "令 $F = 0$ ，则 $400{,}000y = 3{,}200{,}000$ ，解得 $y = 8$ 。"

- 因此，该能力是隐含但必要的解题组成部分。

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5. 第13项：利用对情境的理解指导或加速数学求解过程

> *"Use an understanding of the context to guide or expedite the mathematical solving process, e.g., working to a context-appropriate level of accuracy"*

- 中等相关：情境知识帮助学生：
  - 理解"收回成本"即 $F = 0$ （而非 $F > 0$ ）；
  - 知道年份必须为整数，且取最小满足值（故 $y = 8$ ，非 7.999...）；
  - 无需保留小数或考虑连续时间模型。
- 这种情境驱动的简化使计算更高效、答案更合理。

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6. 第1项：进行简单计算

> *"Perform a simple calculation"*

- 弱相关但存在：最终只需一次除法（3.2M ÷ 0.4M = 8）。
- 虽然计算本身简单，但它是多步过程的最后环节，属于基础执行能力。